[ISOM2010] Introduction to Information Systems

Digital Economy III & IV

Digital Goods and Digital Markets

S4-5

Prof. Kim, Yongsuk

DIGITAL GOODS

What are digital goods

- **Digital goods = Information Goods**
- **The "Bit" Concept:** Anything that can be expressed as a string of 1s and 0s.
- **From Atoms to Bits:**
  - Physical Goods (Atoms): Have mass, take up space, and cost money to move.
  - Digital Goods (Bits): Weightless, move at the speed of light, and occupy no physical space.
- **Examples:** images, videos, music, software products, virtual assets (e.g., in-game skins), digital currencies

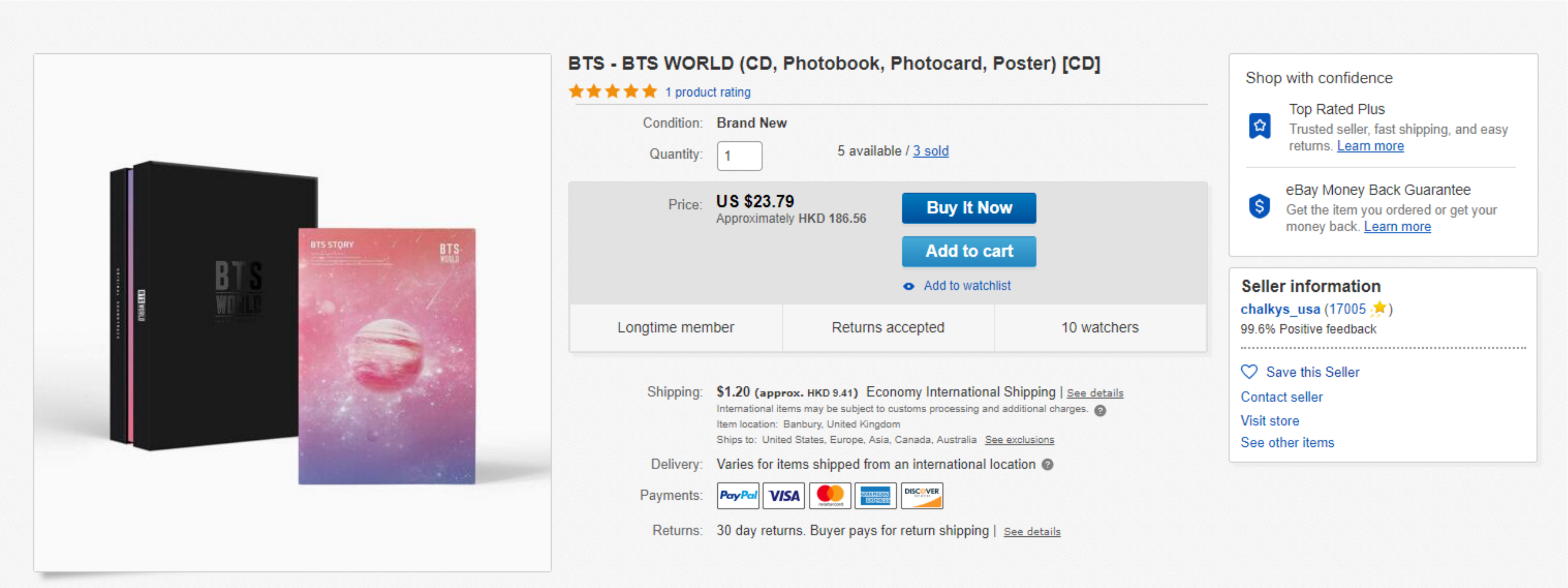
The New York Times

DIGITAL GOODS

Digital: Easy to copy, hard to protect

- **First Sale Doctrine**
  - If a copyrighted work is purchased, the purchaser has the right to transfer that copy.

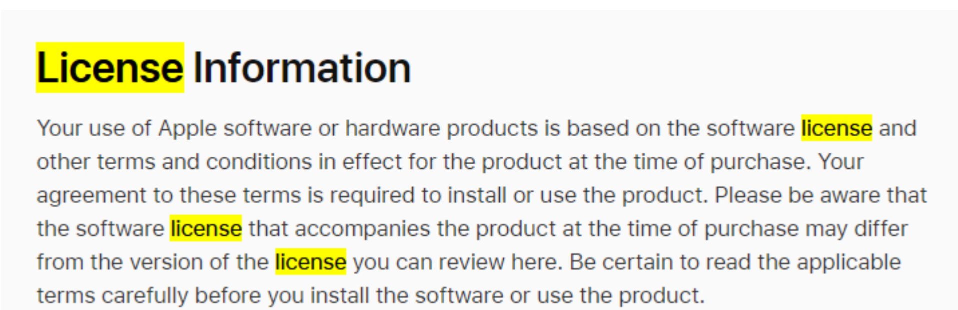
    → Enables individuals to sell their legally purchased books or CDs to others.



Digital Goods

Digital: Easy to copy, hard to protect

- **Licensed, not sold**
  - Most of digital goods are supplied under license, instead of being sold.
    - Consumers can easily give up their fist sale doctrine rights by agreeing to certain contract.
    - Someone who purchases a software license is **not the "owner of a particular copy,"** they are an **"owner of a license to use a copy"** of the software.
  - **DRM(Digital Rights Management)** enables to control the use, modification, and distribution of copyrighted works.

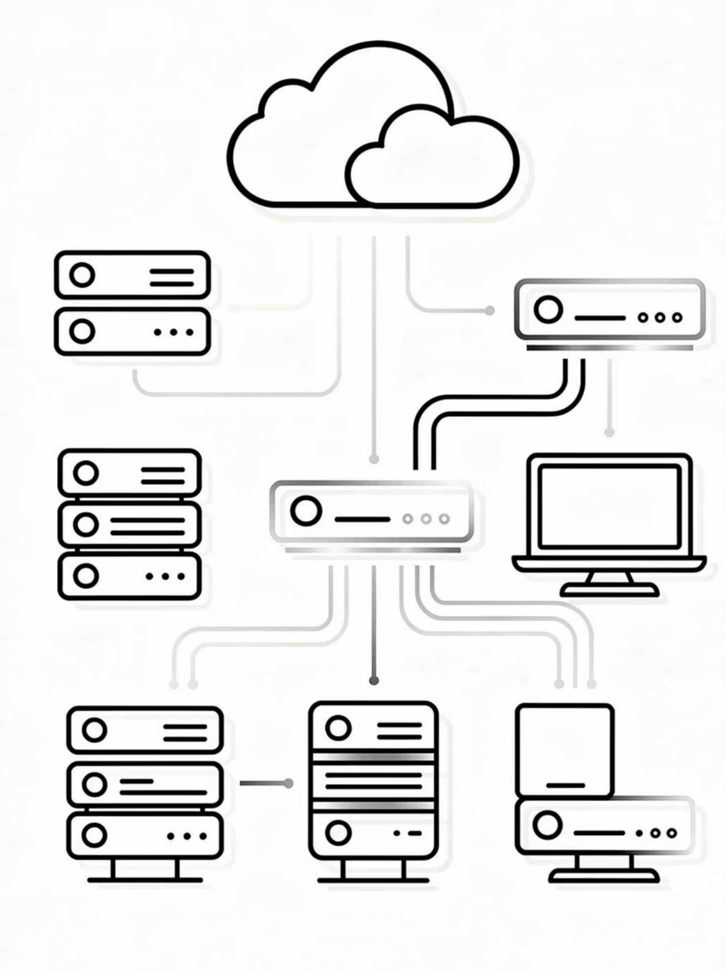


License Information

Your use of Apple software or hardware products is based on the software license and other terms and conditions in effect for the product at the time of purchase. Your agreement to these terms is required to install or use the product. Please be aware that the software license that accompanies the product at the time of purchase may differ from the version of the license you can review here. Be certain to read the applicable terms carefully before you install the software or use the product.

Eight Unique Properties of Digital Goods

Understanding how digital products fundamentally differ from physical goods through the lens of Amazon vs. traditional book stores



1. **Nearly Zero Marginal Cost**
2. **Non-Rivalry**
3. **In-destructibility**
4. **Ubiquity & Global Reach**
5. **Richness (Information Density)**
6. **Interactivity**
7. **Personalization / Customization**
8. **Social (Network Effects)**

the cost added by producing one additional unit of a product or service

Property 1

Nearly Zero Marginal Cost (with High Fixed Cost)

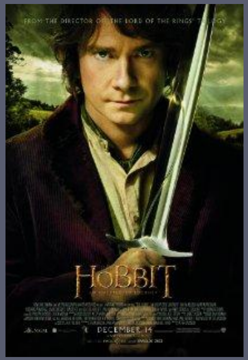
Concept: Digital goods require a massive initial investment to create the first unit (High Fixed Cost), but once created, replicating it costs almost nothing (Nearly Zero Marginal Cost).

Physical Goods: To sell another physical book, a book store must pay for more paper, ink, binding, and shipping for every single copy.

Digital Goods: After the author and publisher spend months creating one E-book file, delivering that file to the 1,000,001st customer via Amazon costs virtually $0.

The "Why"

This allows digital companies to scale infinitely and achieve massive profit margins that physical stores cannot match.



Digital goods shift the business focus from variable costs to fixed costs, allowing for infinite scaling. While physical goods scale linearly (more units = more costs), digital goods scale exponentially (more units = more profit margin).

THE REAL COST *of* MUSIC [3]

Online music services don't incur packaging, distribution, and retail fees — and they should charge accordingly.

CREATION COSTS

Artist	$1.50
Marketing and Profit	$5.00
Publishing	$0.96
	$7.46

→ Divided by 12 tracks = $0.62/track + 17¢ online delivery cost ↓ **$0.79/song**

\+

PRODUCTION COSTS

Packaging	$0.75
Distribution	$2.00
Retail markup	$5.00
	$7.76

↓

$15.21/CD

Source: http://changethis.com/manifesto/10.LongTail/pdf/10.LongTail.pdf

Property 2

Non-Rivalry

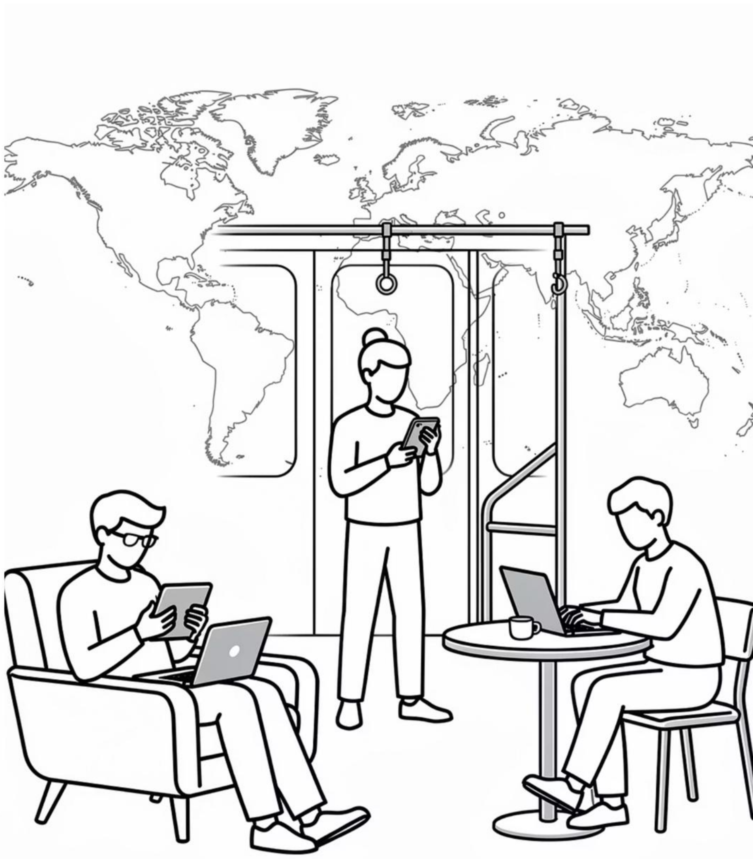
Concept: Digital goods are not "consumed" or "used up." One person's use does not reduce the availability for others.

Physical Goods

Physical books are Rivalrous. If a bookstore has only one copy of a textbook and you buy it, no one else can have it until a new shipment arrives.

Digital Goods

E-books are Non-Rivalrous. An entire class of 200 students can download and read the exact same file on their Kindles at the same time without "running out of stock."

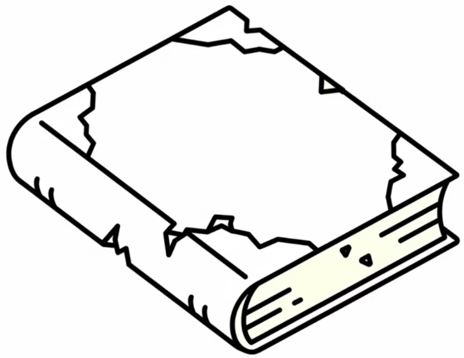


Digital resources are not consumed or used up by a single user, enabling simultaneous utility. Unlike physical goods which are "zero-sum" (if I have it, you don't), digital goods are "infinite-sum."

Property 3

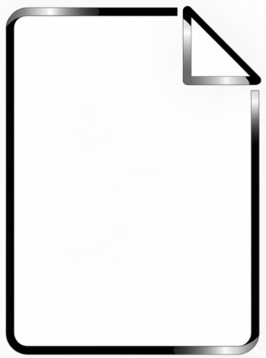
In-destructibility

Concept: Unlike physical matter, digital data does not suffer from entropy or physical wear and tear.



Physical Goods

Physical books are fragile. Pages turn yellow over time, covers get torn, and coffee spills can ruin the product forever.



Digital Goods

A digital file from 2006 is identical in quality to one from 2026. No pixels fade, and no data "breaks" from frequent use. It stays "brand new" for eternity.

Digital goods exist as pure data, immune to the physical entropy that degrades all tangible objects. Physical goods are depreciating assets that wear out, while digital goods are "forever" assets that maintain 100% quality.

Reverse Aging / Depreciation

In fact, through **updates**, digital goods can often improve over time instead of losing value.

	Physical Goods (Atoms)	Digital Goods (Bits)
Value over Time	**Value Decreases:** A 5-year-old car is usually worse than a brand-new one due to mechanical decay.	**Value Increases (Optional):** Through regular updates, a digital product (like an App or Tesla software) can actually get **better and smarter** than when you first bought it.

Q. Implications for the business model?

Property 4

Ubiquity & Global Reach

Concept: Digital goods exist in a borderless "Marketspace" accessible anytime, anywhere, removing the friction of distance.

1	2
Physical Goods You must physically travel to a bookstore during its operating hours. If you are in Hong Kong, you cannot easily browse a small bookstore in London.	**Digital Goods** Amazon is in your pocket 24/7. Whether you are at home or traveling, you can purchase and start reading a book in seconds, transcending geography and time.

Digital goods eliminate the geographical friction of trade, making themselves omnipresent. Physical goods require a physical "Marketplace," but digital goods exist in a borderless "Marketspace."

Digital Good Insight: The Product That Travels at the Speed of Light

Feature	Physical Good (Atoms)	Digital Good (Bits)	DG Strategic Insight
Product Existence	**Place-Bound:** The product is a prisoner of a shelf. It stays where it is put.	**Ubiquitous:** The product exists in the "Cloud." It can be summoned anywhere, anytime.	**From Shelf to Cloud:** The product is no longer a "thing" you go to see; it is a "service" that follows you.
Delivery Friction	**Geographical Limit:** A physical book's reach is limited by the cost of shipping and distance.	**Global Reach:** A digital file has zero "weight." It reaches a user in HK or London instantly.	**The Death of Distance:** The physical location of the creator no longer limits the reach of the product.
Availability	**Synchronous:** You can only use the product if you have the physical object in hand.	**Always-On:** The product is "Always-ready." It doesn't sleep, and it doesn't have "opening hours."	**Constant Utility:** Digital goods provide value 24/7 without requiring a physical middleman.

Property 5

Richness (Information Density)

Concept: Digital goods can layer vast amounts of multi-dimensional information without increasing the physical size or weight of the product.

Physical Goods

A book is limited to the ink on its paper. To get more info, you'd need a bigger, heavier book.

Digital Goods

A single E-book can include "X-Ray" character bios, instant dictionary lookups, and embedded video interviews with the author—all within a device that weighs less than a single paperback.

Physical goods are limited by their dimensions, but digital goods are limited only by data storage and bandwidth. DGs can also combine Text, Audio, Video, and Interactive elements, whereas PGs are mostly limited to static text and images.

Category	Physical Goods (Atoms)	Digital Goods (Bits)
Navigation	**Paper Maps:** Limited to printed roads. To see more detail, you need a larger, heavier map.	**Google Maps:** Unlimited layers. Includes live traffic, 3D street views, and restaurant reviews in one app.
Education	**Textbooks:** Mostly static text and images. You need a separate CD or QR code to hear audio.	**Duolingo:** All-in-one. Combines text, native audio, animations, and instant voice correction.
Entertainment	**DVDs/Blu-rays:** Fixed content. You must change the physical disc to watch a different movie.	**Netflix:** Deep content. Switch languages instantly and see actor bios or related shows without moving.

Property 6

Interactivity (human–product, human-human)

Interactivity

Concept: Digital goods create a two-way dialogue; the product reacts to the user's behavior and input.

Physical Goods: A physical book is a "passive" object. It doesn't know if you are confused by a chapter or if you've stopped reading halfway through.

Digital Goods: Amazon/Kindle tracks your reading speed, remembers your highlights, and can even offer interactive quizzes or "smart" summaries based on which parts of the book you spent the most time on.

Digital goods transform the relationship from passive consumption to active engagement. A physical good is a "dead" object, whereas a digital good is a reactive system that learns from user input and also facilitates p2p interactions.

Property 7

Personalization

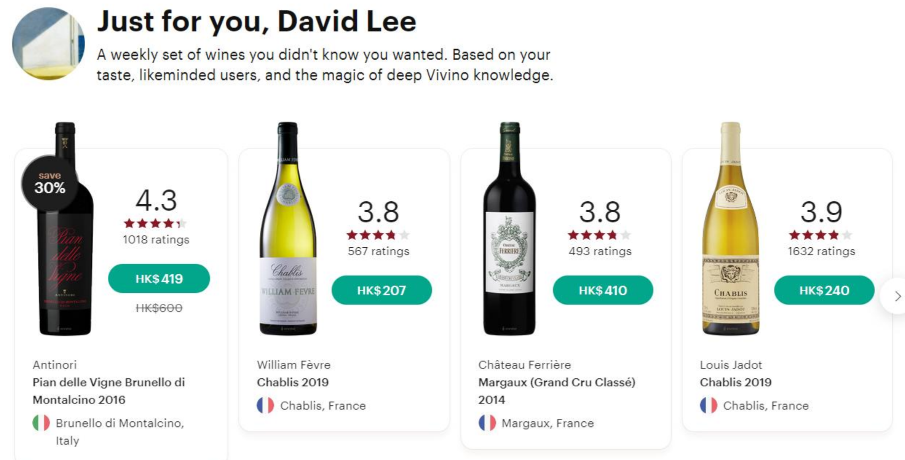
Personalization / Customization

Concept: Digital goods can be tailored to the unique preferences of a single individual at an algorithmic scale.

Physical Goods: Every customer walks into a bookstore and sees the same "Staff Picks" shelf. Customizing a physical book for one person is too expensive for mass retail.

Digital Goods: No two Amazon homepages are the same. Based on your browsing history, Amazon builds a "Segment of One" storefront that only shows books you are likely to love.

Digital goods allow for mass-individualization, where every user receives a unique version of the product. Customizing physical goods is slow and artisanal; customizing digital goods is algorithmic and instantaneous.



Property 8

Social (Network Effects)

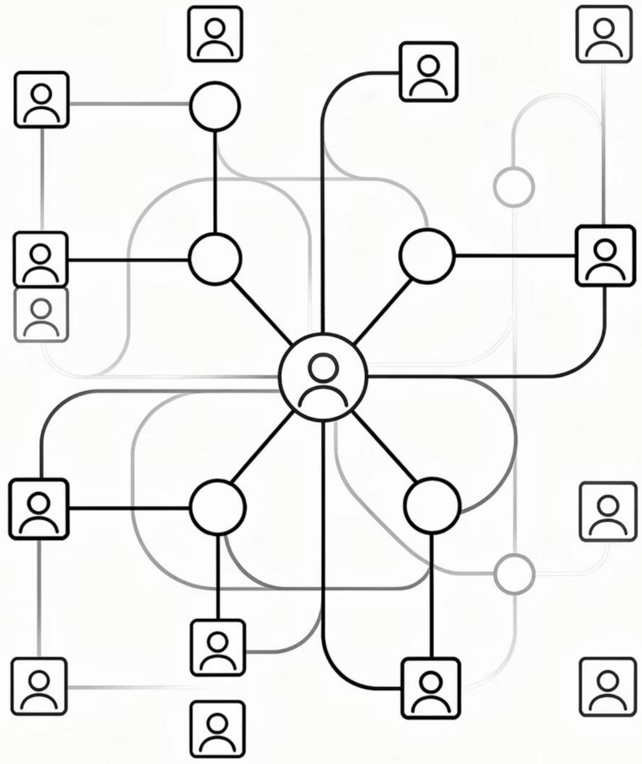
Concept: The value of a digital good increases as the community of users grows, as they contribute data that improves the experience for everyone.

Physical Goods

If an offline bookstore is too crowded, the experience gets worse. Your purchase doesn't make the book better for the next person.

Digital Goods

On Amazon, your reviews, ratings, and "Customers who bought this also bought..." signals create a feedback loop. The more people use the platform, the smarter the recommendations become for everyone.



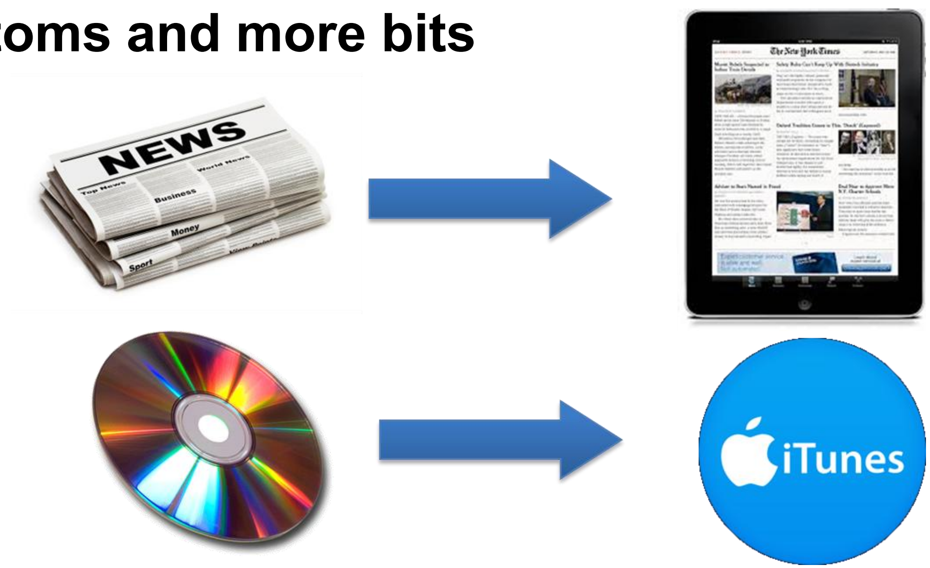
The value of a digital good is a function of its community; more users create more data, which in turn creates more value. Physical value is set at the point of sale, but digital value grows dynamically as the network expands.

DIGITAL GOODS

All Goods are combinations of both Atoms + Bits

- **A= atoms, B=bits, P=price**
- **Beer: {A, B, P}**
- **Music: {A, B, P}**

- **As time goes by in more and more product categories, new products have fewer atoms and more bits**



Digital Goods

Impact of New Technologies

Books
Music
Movies
Financial Reports

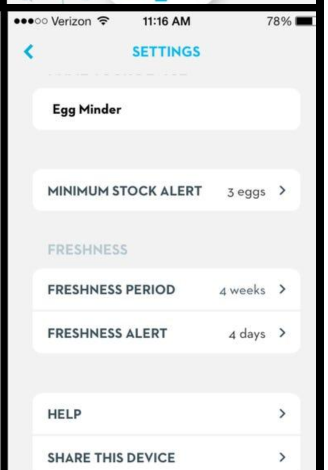
Cars
Microwave ovens
Washing Machines

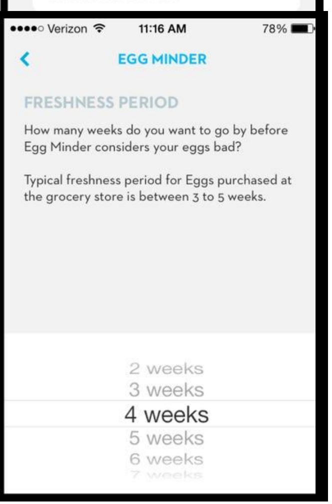
Digital Goods — Physical Goods

Impact of New Technologies

- **Move towards Hybrid Digital/Information Products**
  - Examples: Smart Car, Smart Refrigerator
  - https://www.youtube.com/watch?v=z6LtA2cSvUU

- **Internet of Things (IoT)**
  - Objects, animals or people are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction
  - IBM Internet of Things https://www.youtube.com/watch?v=8R0CPFIU14M

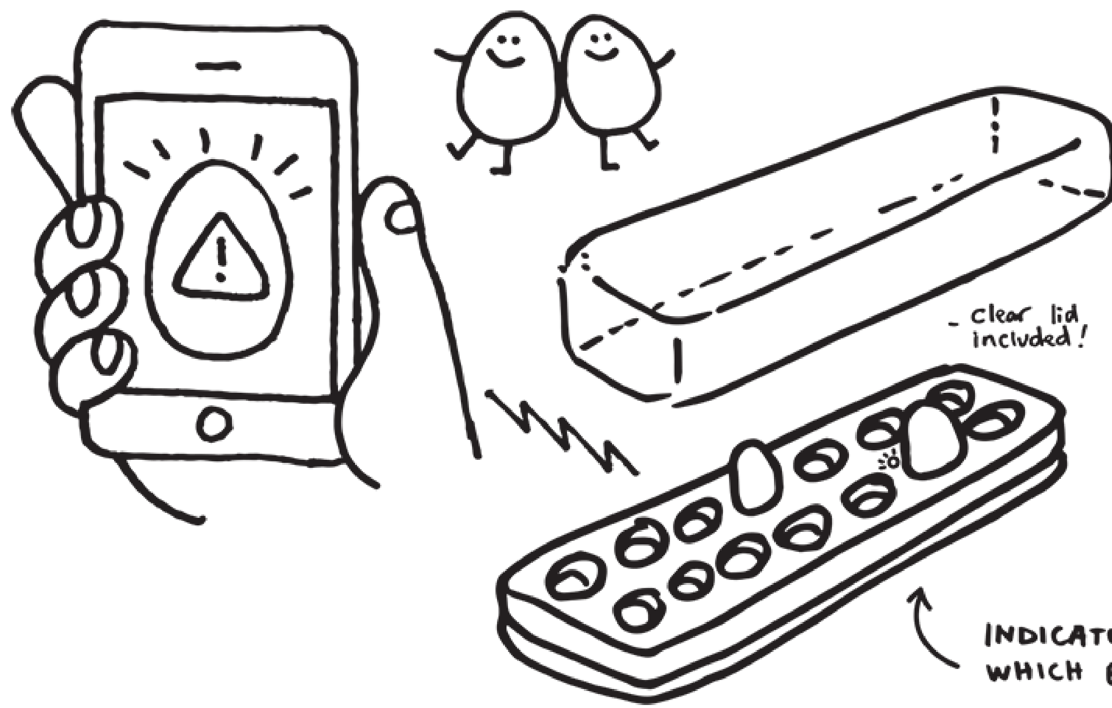




FIND OUT...

WHEN YOU'RE LOW ON EGGS AND
WHEN YOUR EGGS ARE GOING BAD

As seen on The Tonight Show with Jay Leno:



inventor:
RAFAEL HWANG
Glendale, AZ
+2383 influencers

INDICATOR LIGHT SHOWS
WHICH EGG IS THE OLDEST

Impact of New Technologies

Books
Music
Movies
Financial Reports

Cars
Microwave ovens
Washing Machines

Digital Goods

Physical Goods

Impact of New Technologies

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Tesla: The Ultimate Hybrid Product

Software-Defined Vehicle (SDV)

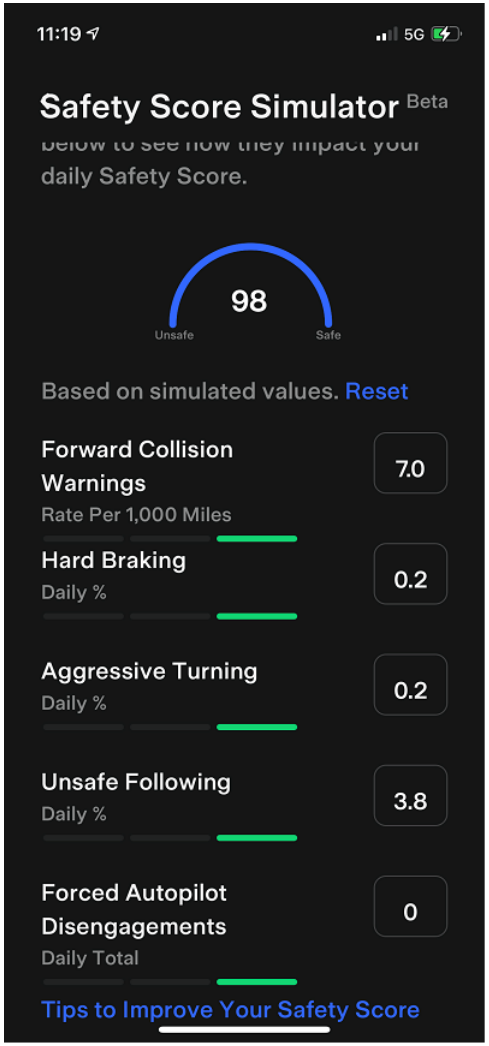
- **The Shift:** Historically, cars were "dumb" mechanical machines. Tesla redefined the car as a **“Computer on Wheels.”**

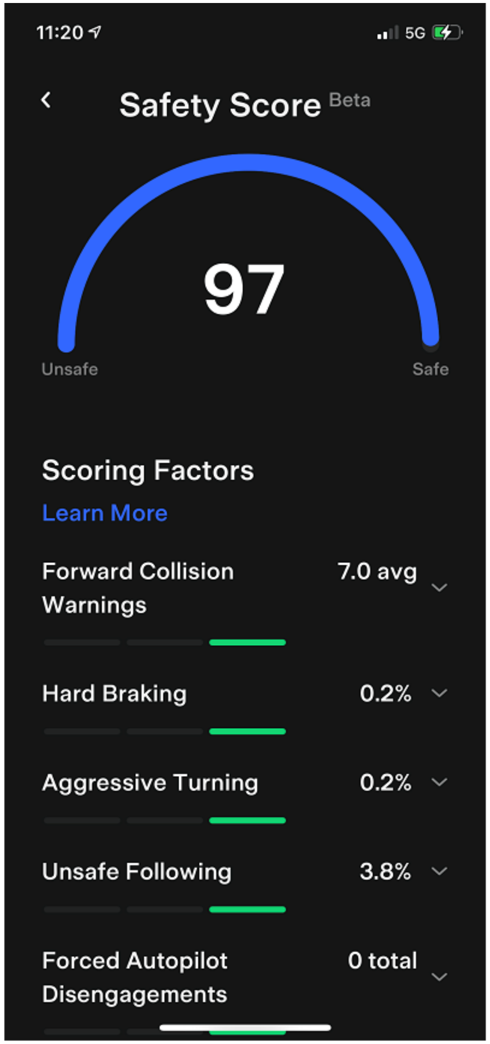
- **Atoms to Bits:** While the chassis and tires are "Atoms," the core value of the car—its performance, safety features, and battery management—is controlled by **"Bits" (Software).**

Tesla: The Ultimate Hybrid Product

Characteristic	Traditional Car (Physical)	Tesla (Digital/Hybrid)
Nearly Zero Marginal Cost	Adding speed requires new parts and labor for every car.	**Digital Upgrades:** Selling a speed boost to one more user costs Tesla almost $0.
Non-Rivalry	One engine, one user. Limited by physical parts.	**Shared Intelligence:** Everyone uses the same AI "brain" simultaneously.
In-destructibility	Parts wear out and get old the moment you drive away.	**Forever New:** Software updates fix bugs and add features so the car gets better over time.
Ubiquity & Global Reach	You must visit a physical shop to buy or fix things.	**App Control:** You can track, lock, or upgrade your car from anywhere, 24/7.
Richness	A paper manual and a few simple needles on the dash.	**Data Hub:** A giant screen filled with high-tech maps and live car data.
Interactivity	A "dead" machine. It doesn't talk back or learn.	**Two-way Dialogue:** The car reacts to your driving and gives you real-time feedback.
Personalization	Fixed features. "What you buy is what you get."	**Software Profiles:** The car's behavior and UI change instantly for every driver.
Social	One person's car has nothing to do with another's.	**Crowdsourced AI:** Every mile you drive helps make *all* Tesla cars smarter.

Tesla: The Ultimate Hybrid Product





Key Concepts in Digital Markets

Digital Market

a virtual environment where commercial transactions occur electronically via the internet, rather than in a physical "Marketplace".

Digital Markets vs. Traditional Markets

	DIGITAL MARKETS	TRADITIONAL MARKETS
Information asymmetry	Asymmetry reduced	Asymmetry high
Search costs	Low	High
Transaction costs	Low (sometimes virtually nothing)	High (time, travel)
Delayed gratification	High (or lower in the case of a digital good)	Lower: purchase now
Menu costs	Low	High
Dynamic pricing	Low cost, instant	High cost, delayed
Price discrimination	Low cost, instant	High cost, delayed
Market segmentation	Low cost, moderate precision	High cost, less precision
Switching costs	Higher/lower (depending on product characteristics)	High
Network effects	Strong	Weaker
Disintermediation	More possible/likely	Less possible/unlikely

Source: Laudon and Laudon (2018)

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Information Asymmetry

- A situation where the relative bargaining power of two parties in a transaction is determined by one party of the transaction possessing more information essential to the transaction than the other party.

INFORMATION ASYMMETRY

Information Asymmetry

Canon PowerShot G7 X Mark II Digital Camera w/1 Inch Sensor and tilt LCD Screen - Wi-Fi & NFC Enabled (Black)

by Canon

253 customer reviews | 333 answered questions

Amazon's Choice for "canon g7x mark ii"

List Price: ~~$699.00~~

Price: $649.00 prime | FREE One-Day

You Save: $50.00 (7%)

Style: **Camera Only**

Camera Only	Memory Card Bundle	Video Creator Kit	w/ Canon Battery Pack
$649.00 prime	$664.19 prime	$699.00 prime	- -

253

4.4 out of 5 stars

5 star
4 star
3 star
2 star
1 star

See all 25

Rated b

Picture qu

4.6 out of

Read re

picture

lens

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Top customer reviews

Uriah Werner

This camera is amazing for its size.

July 1, 2016

Style: Camera Only | Configuration: Base | **Verified Purchase**

Amazing camera. I'm a professional photographer and I am extremely impressed with the picture quality. I also love all the manual controls and the ease-of-use. I shoot weddings on a Canon 5D Mark III and a Canon 6D. In a pinch I could easily use this camera if I needed to.

The picture I have attached is a 15 second exposure with my kids running around with solar lights. This shot is un-edited and was shot in raw. The only thing I did with the picture was transfer it via Wi-Fi to my cell phone using the Canon picture app.

683 people found this helpful

Helpful | Not Helpful | Comment | Report abuse

Matthew S

Nearly A Perfect Camera. Nearly a Complete DSLR Replacement

June 17, 2016

Style: Camera Only | Configuration: Base

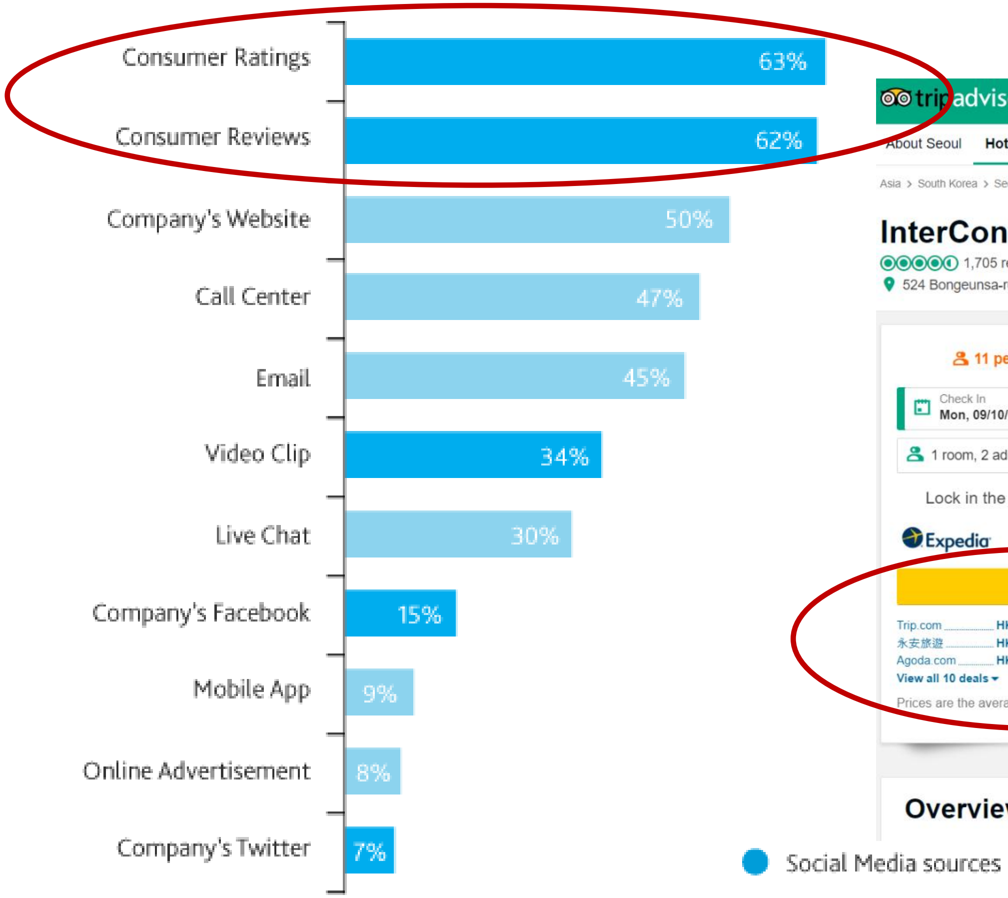
I am a hobby shooter, but prefer quality and do a lot of YouTube videos. For background, I've had a variety of different camcorders, point-and-shoots, and DSLRs: Sony RX100 III,

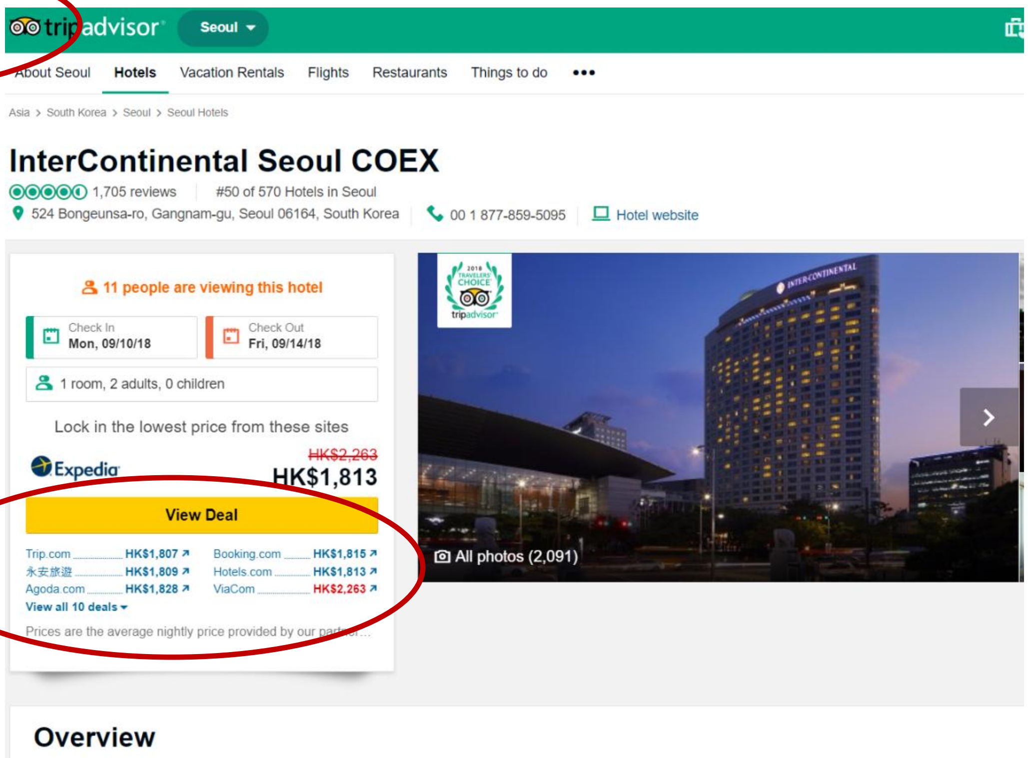
ttps://www.amazon.com/Canon-PowerShot-Digital-Camera-Sensor/product-reviews/B01BV14OXA/ref=cm_cr_dp_d_hist_3?ie=UTF8&filterByStar=three_star&reviewerType=all_reviews#reviews-filter-bar

⇩ INFORMATION ASYMMETRY

Which is the preferred source for product and service information?

% Social Media Users





Read as: 63% of social media users list Consumer Ratings at their preferred source for information about products/services.

Source: NM Incite

Source: http://blog.nielsen.com/nielsenwire/consumer/how-social-media-impacts-brand-marketing/

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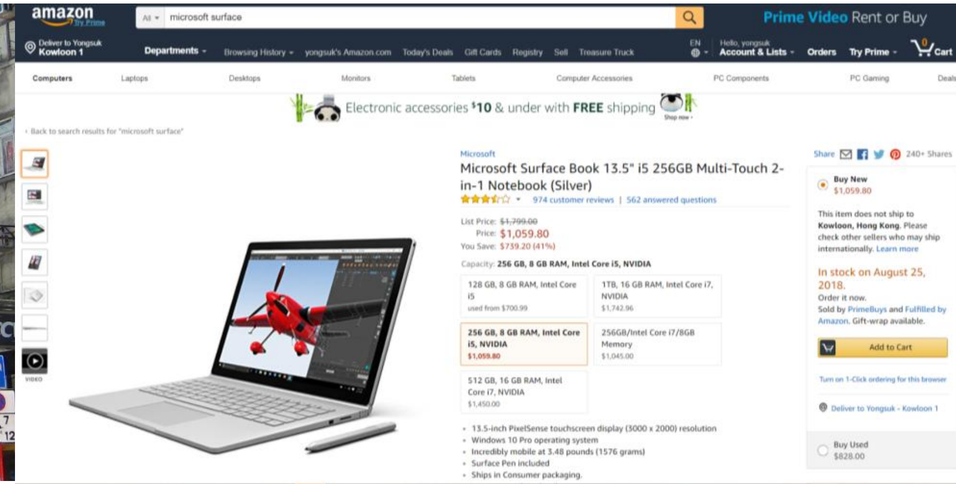
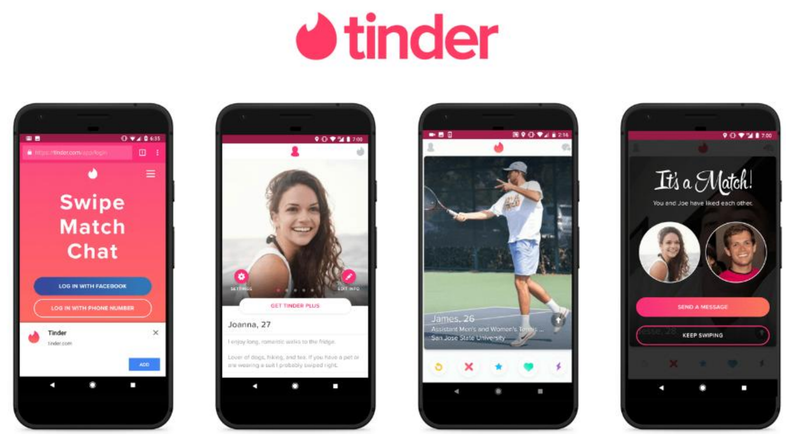
Transaction Costs

- Costs associated with completing a transaction
  - Search and information costs: to find the right supplier
  - Bargaining and decision costs: to purchase components
  - Policing and enforcement costs: to monitor quality

Transaction Costs

Search Cost

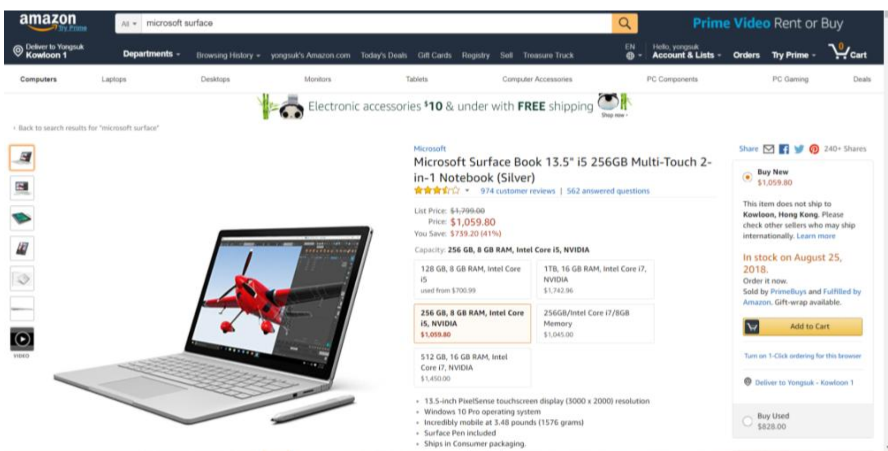




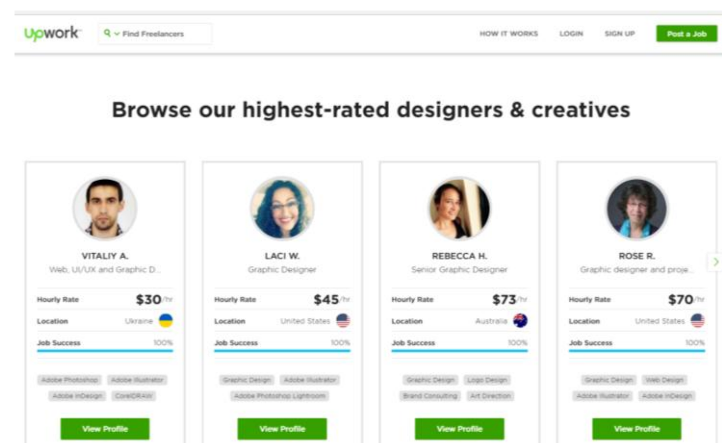
Transaction Costs

TYPES OF COST

- SEARCH AND INFORMATION
- BARGAINING AND DECISION
- POLICING AND ENFORCEMENT



Online Product Marketplace



Online Service Marketplace

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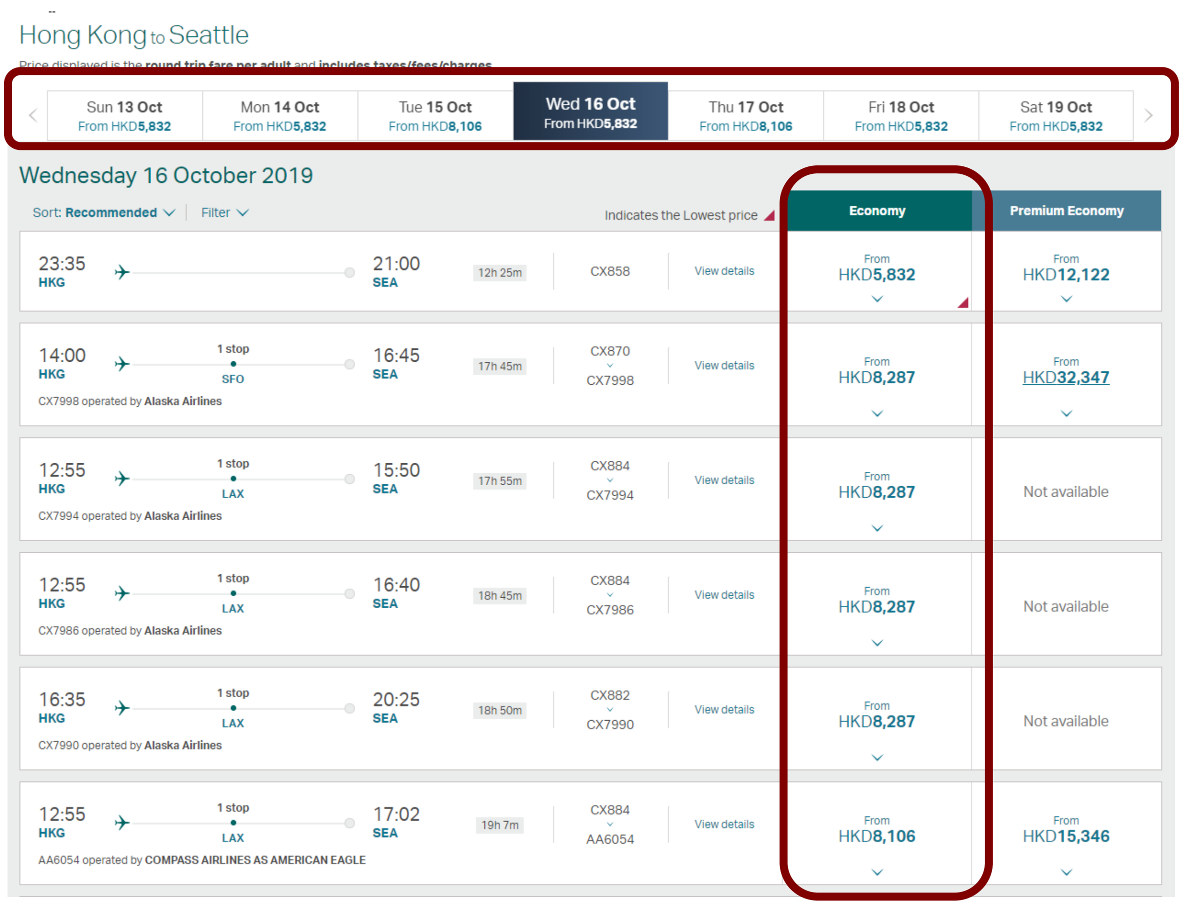
Dynamic Pricing

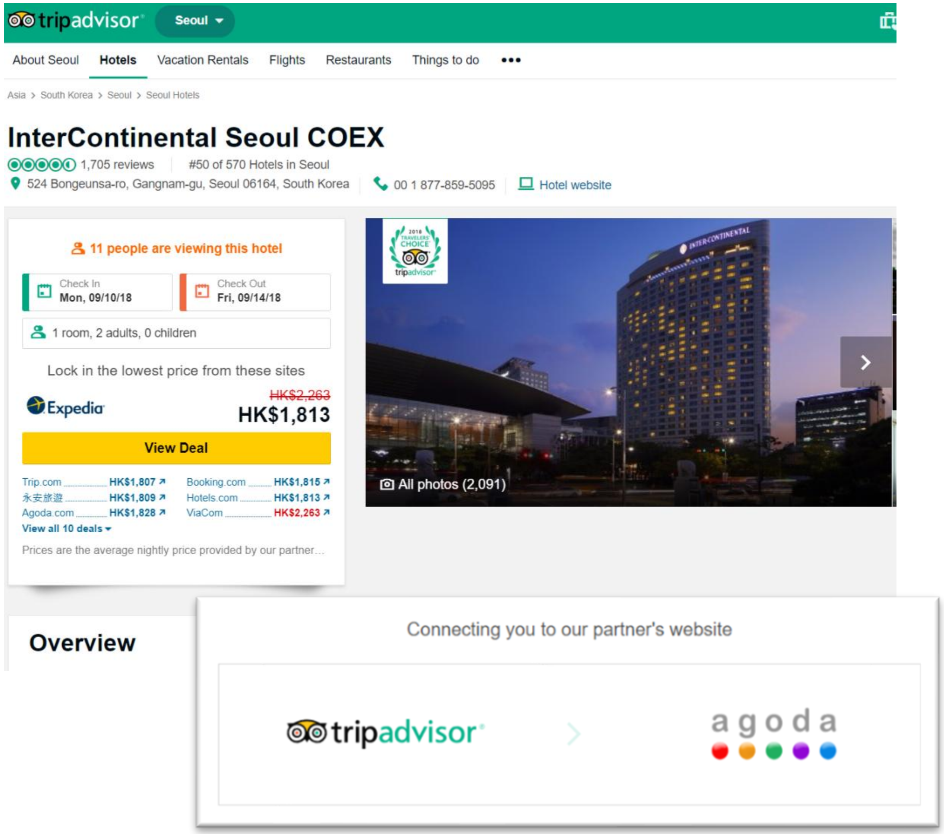
- **Pricing of items based on real-time interactions between buyers and sellers that determine what an item is worth at any particular moment**



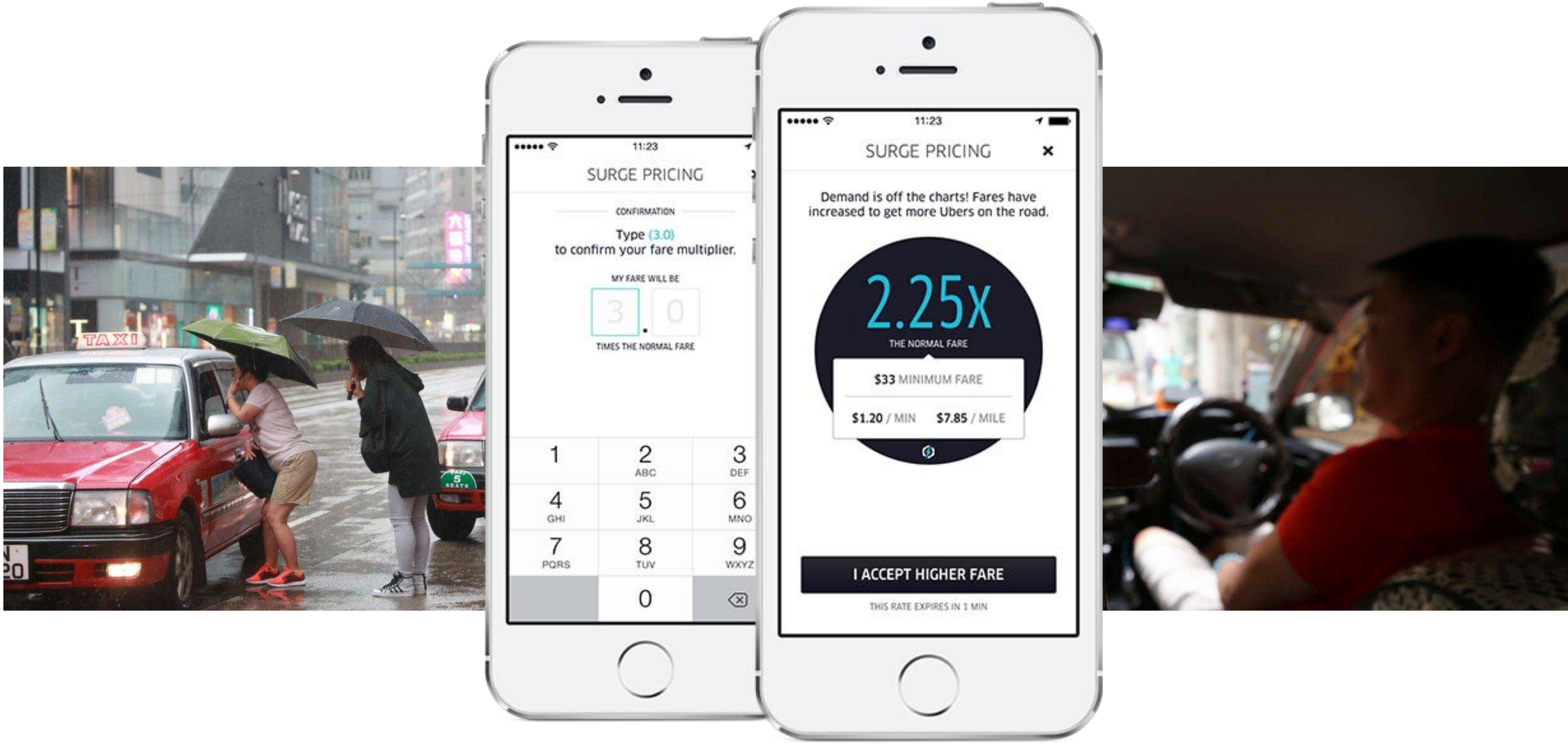
Factors affecting Dynamic Pricing

Dynamic Pricing

- **Factors that determine a dynamic price**
  - Time of day or night
  - Customer location
  - Day of the week
  - Level of demand
  - Competitor pricing
  - Method / stage of purchase



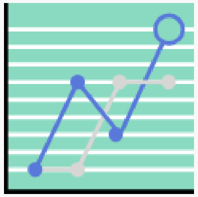
Dynamic Pricing



Dynamic Pricing: Uber Case

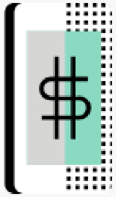
What is surge pricing?

Here's how it works



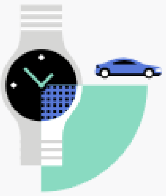
Demand for rides increases

There are times when so many people are requesting rides that there aren't enough cars on the road to help take them all. Bad weather, rush hour, and special events, for instance, may cause unusually large numbers of people to want to ride Uber all at the same time.



Prices go up

In these cases of very high demand, fares may increase to help ensure those who need a ride can get one. This system is called surge pricing, and it lets us continue to be a reliable choice.

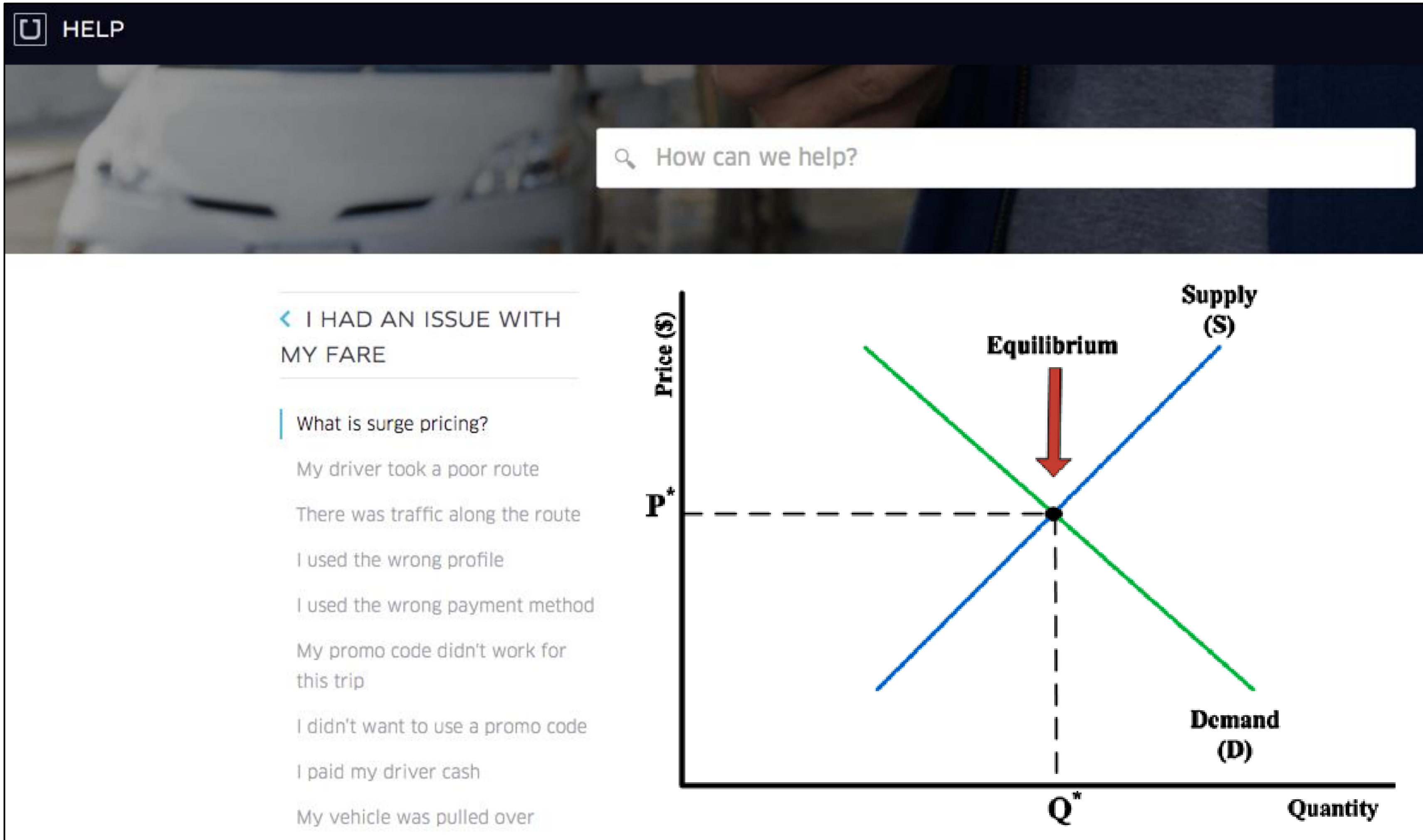


Riders pay more or wait

Whenever we raise rates due to surge pricing, we let riders know in the app. Some riders will choose to pay, while some will choose to wait a few minutes to see if the rates go back down to normal.

DYNAMIC PRICING: UBER CASE

Dynamic Pricing



Source: http://www.bloombergview.com/articles/2016-02-08/uber-s-surge-pricing-applied-to-smoothies

Dynamic Pricing: Uber Case

Dynamic Pricing

Uber's Surge Pricing Near Siege in Sydney Sparks Outrage

Dec.15.2014 / 2:46 PM ET / Updated Dec.15.2014 / 9:05 PM ET

Uber Slammed for Surge Prices After New York City Bombing

September 19, 2016

Uber slammed for surge pricing during London attacks

By Mark Moore

June 4, 2017 | 9:37pm | Updated

Uber offers free rides, Lyft suspends surge pricing in Las Vegas after deadly mass shooting

Mashable | RACHEL KRAUS

Oct 3rd 2017 10:34AM

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Source: Laudon and Laudon (2018)

Price Discrimination

Price Discrimination

- **Selling the same goods, or nearly the same goods, to different targeted groups at different prices**

Price Discrimination – Types

- ***First Degree (Perfect)***
  - Charging the maximum price consumers are willing to pay.
  - Examples: personalized pricing, negotiating the price of a used car

- ***Second Degree***
  - Charging different prices depending on the quantity purchased or version of the product chosen.
  - Offer different price-product bundles that are attractive to the "right" customers but not others
  - Examples: quantity discount, different versions of products

- ***Third Degree***
  - Charging different prices depending on a particular market segment.
  - Examples: student discounts, senior discounts for public transportation, charging different prices in a different geographic market

Price Discrimination

THE WALL STREET JOURNAL.

U.S. EDITION Monday, December 24, 2012

WHAT THEY KNOW | December 24, 2012

Websites Vary Prices, Deals Based on Users' Information

Article | Interactive Graphics | Comments (134)

Email Print Save

By JENNIFER VALENTINO-DEVRIES, JEREMY SINGER-VINE and ASHKAN SOLTANI

It was the same Swingline stapler, on the same Staples.com website. But for Kim Wamble, the price was $15.79, while the price on Trude Frizzell's screen, just a few miles away, was $14.29.

A key difference: where Staples seemed to think they were located.

A Wall Street Journal investigation found that the Staples Inc. website displays different prices to people after estimating their locations. More than that, Staples appeared to consider the person's distance from a rival brick-and-mortar store, either OfficeMax Inc. or Office Depot Inc. If rival stores were within 20 miles or so, Staples.com usually showed a discounted price.

Source: http://online.wsj.com/article_email/SB10001424127887323777204578189391813881534-lMyQjAxMTAyMDIwMzEyNDMyWj.html

PRICE DISCRIMINATION

CBS 46

NEWS WEATHER TRAFFIC VIDEOS 46 INVESTIGATES PLUGGED IN

Facebook Twitter Print Email More 326

Mac users might pay more when shopping online

Posted: Jan 24, 2015 12:06 AM
Updated: Apr 21, 2015 3:15 AM

ATLANTA (CBS46) - You could be paying higher online prices depending on your internet history and what device you use to shop.

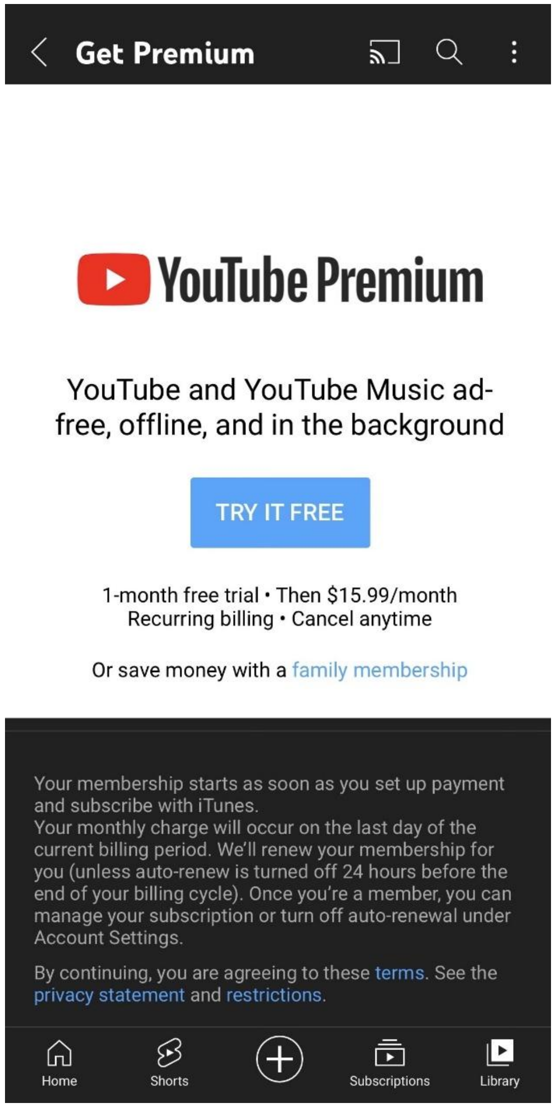
A CBS46 investigation found online retailers are manipulating prices to charge you more. If you're a loyal Apple user, the price may automatically go up.

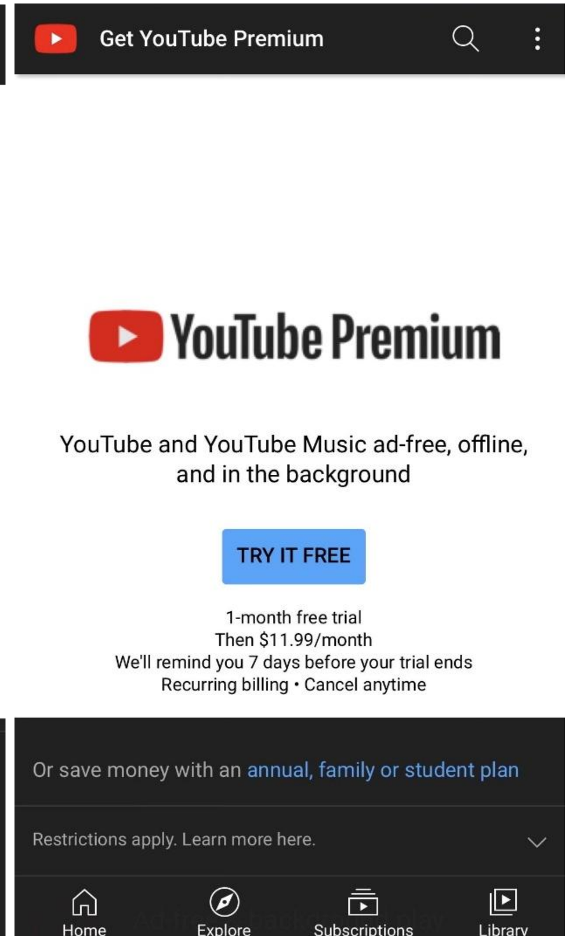
It may not happen every time, but CBS46 found instances where a search for the same product on the same website produced not only different results, but different prices and it's perfectly legal. When you pull an item from a store shelf, the next person to grab the same thing pays the same price. That's not always the case with online retailers.

"Some are more sophisticated than others so if you go and shop at a particular retailer and you're seeing the exact same price no matter what piece of hardware, no matter what browser you use, they're probably a little less sophisticated. They haven't quite gotten to the point where they can customize those models," said Jennifer Lewis Priestley, Ph.D., professor of applied statistics and data

Price Discrimination

iOS





android

Factors affecting Price Discrimination

Price Discrimination

- **Selling the same goods, or nearly the same goods, to different targeted groups at different prices**

- **Ways to separate customers (highly personal!)**
  - **Geography (zip code)**
  - **Income**
  - **Gender**
  - **Age**
  - **Time**
  - **Race**
  - **Language**
  - **Devices used to log on**
  - **Even OS/browser in use**
  - **Shopping behaviors**

Air ticket Fares: Dynamic pricing and price differentiation

What affects airfares?

- Supply and demand of seats
- Peak travel seasons
- When tickets are purchased

+ based on customer profile

- Birthday, address, martial status, nationality, gender, etc.
- Travel history, frequent flier status
- Trip purpose: Leisure travelers vs. business travelers

Each of us will be asked to pay a different fare based on who we are??? (individual discrimination? Information asymmetry going up again?)

Price Discrimination

Crackdown on retailers that use Big Brother tactics to identify shoppers with expensive Apple computers - so they can quote higher prices

- **Business Secretary Kwasi Kwarteng has plans to stop consumers losing out**
- **Online retailers are charging more for customers who use expensive computers**
- **Retailers are able to gather large amounts of data from online shoppers**
- **Plans to strengthen the role of the Competition Authority are being considered**

By ANNA MIKHAILOVA and MAX AITCHISON FOR THE MAIL ON SUNDAY
PUBLISHED: 22:00 GMT, 17 July 2021 | **UPDATED:** 23:12 GMT, 17 July 2021

Share 283 shares 375 View comments

The Government is to crack down on greedy online retailers that use Big Brother tactics to identify shoppers with expensive Apple computers so they can quote higher prices.

Officials are examining measures to halt the 'exploitation' of consumers by companies, including airlines, which harvest computer data then use it to tailor prices for individual shoppers.

Business Secretary Kwasi Kwarteng will this week set out plans to tackle rip-off practices, including sneaky fees that are added at checkouts, and forcing companies to switch to an 'opt in' system for subscription renewals.

Tinder Is Phasing Out Higher Prices for Users Ages 29-Plus

In most countries covered by Consumers International's study, people in their 30s and 40s were quoted higher prices than any other age group. In the U.S. their average price was 42.4 percent higher than the price for adults under 30.

People older than 49 saw slightly lower prices on average—much more than 20-somethings, but a bit less than folks in their 30s and 40s.

The age gaps were even higher elsewhere. Dutch Tinder users ages 30 to 49 were quoted more than twice as much on average as younger users in the CI study.

When Tinder Plus launched, age discrimination for the subscription wasn't a secret. In the U.S., swipers older than 30 paid $19.99, while younger users paid $9.99.

Following the rollout, a Tinder user sued the company for age discrimination under California state law. Tinder eventually agreed to settle the class-action lawsuit for $24 million; the terms of the settlement were recently overturned by an appeals court. Tinder didn't admit wrongdoing, but it promised in 2019 to stop the practice for users in California.

https://www.dailymail.co.uk/news/article-9798689/New-rules-stop-online-retailers-charging-Apple-owners-higher-prices.html
https://www.consumerreports.org/consumer-protection/tinder-is-phasing-out-higher-prices-for-users-ages-29-plus-a3881901881/

Data-Driven Personalization in Hospitality: Value vs. Cost

- Today, hotel pricing is evolving from "Demand-based" to **"User-behavior based."** By analyzing granular guest data, hotels can distinguish between "High-Value" and "High-Cost" customers to optimize their Total Profit.

Category	Specific Behavior / Data Points	Business Impact
Value (Ancillary Spend)	Ordering room service, booking spa treatments, minibar usage, casino betting volume, breakfast inclusion.	Increases **LTV (Lifetime Value)** and per-stay revenue beyond the room rate.
Cost (Operational Burden)	Frequency of towel/linen changes, excessive "extra" requests, room condition after checkout, front desk vs. digital check-in.	Increases **CPOR (Cost Per Occupied Room)** and reduces net profit margin.

Case Study: High-Value vs. High-Cost Profiles

- **Case A: The "Low-Impact, High-Yield" Guest (Low Cost, High Value)**
  - **Behavior:** Uses "Do Not Disturb" (DND) for the entire stay, checks in via mobile, but spends heavily at the hotel's Michelin-starred restaurant.
  - **Pricing Strategy:** The hotel may offer this guest a discounted base room rate or complimentary upgrades.

- **Case B: The "High-Maintenance" Guest (High Cost, Low Value)**
  - **Behavior:** Calls the front desk 5 times a day for extra towels/water, leaves the room in a state that requires double the standard cleaning time, and brings outside food instead of using hotel amenities.
  - **Pricing Strategy:** For future stays, the platform might show this guest higher "Base Rates" or exclude them from "exclusive deals.

This shift represents a transition from **Third-degree Price Discrimination** (segmenting by groups like 'Business' vs 'Leisure') to **First-degree Price Discrimination** (individualized pricing).

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Disintermediation	More possible/likely	Less possible/unlikely

Source: Laudon and Laudon (2018)

MARKET SEGMENTATION: NETFLIX CASE

Market Segmentation: Netflix case

There's no such thing as a 'Netflix show'.

Our brand is personalization.

- *Majority of Netflix users consider recommendations with 80% of Netflix views coming from the service's recommendations.*
- *Netflix has set up 1300 recommendation clusters based on users viewing preferences.*
- *Netflix segments its viewers into over 2000 taste groups. Based on the taste group a viewer falls, it dictates the recommendations.*

[How Netflix's Recommendation Engine Works? | by Springboard India | Medium]

Market Segmentation: Netflix case

Feature	Taste Groups (Focus on the User)	Recommendation Clusters (Focus on the Row)
Concept	Segments viewers based on their actual behavior and watching habits.	Algorithmic structures used to organize content into specific "rows" on the UI.
Scale	2,000+ specific groups worldwide.	1,300+ thematic groupings.
Mapping Example	Group A: Fans of "dark, moody crime thrillers with female leads." Group B: Fans of "90s nostalgic sitcoms and colorful cooking competitions."	Cluster 1: “Gritty TV Shows based on Books.” Cluster 2: "Feel-Good Romantic Comedies with a Strong Female Lead."
Key Action	Identifies **Who** the person is (their "flavor").	Decides **What** rows are placed on that person's home screen.
Dynamic Goal	To place a user into a "Segment of One".	To maximize the chance of a click (driving 80% of views).

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Disintermediation		y

Will cover in the sessions on digital platforms!

Source: Laudon and Laudon (2018)

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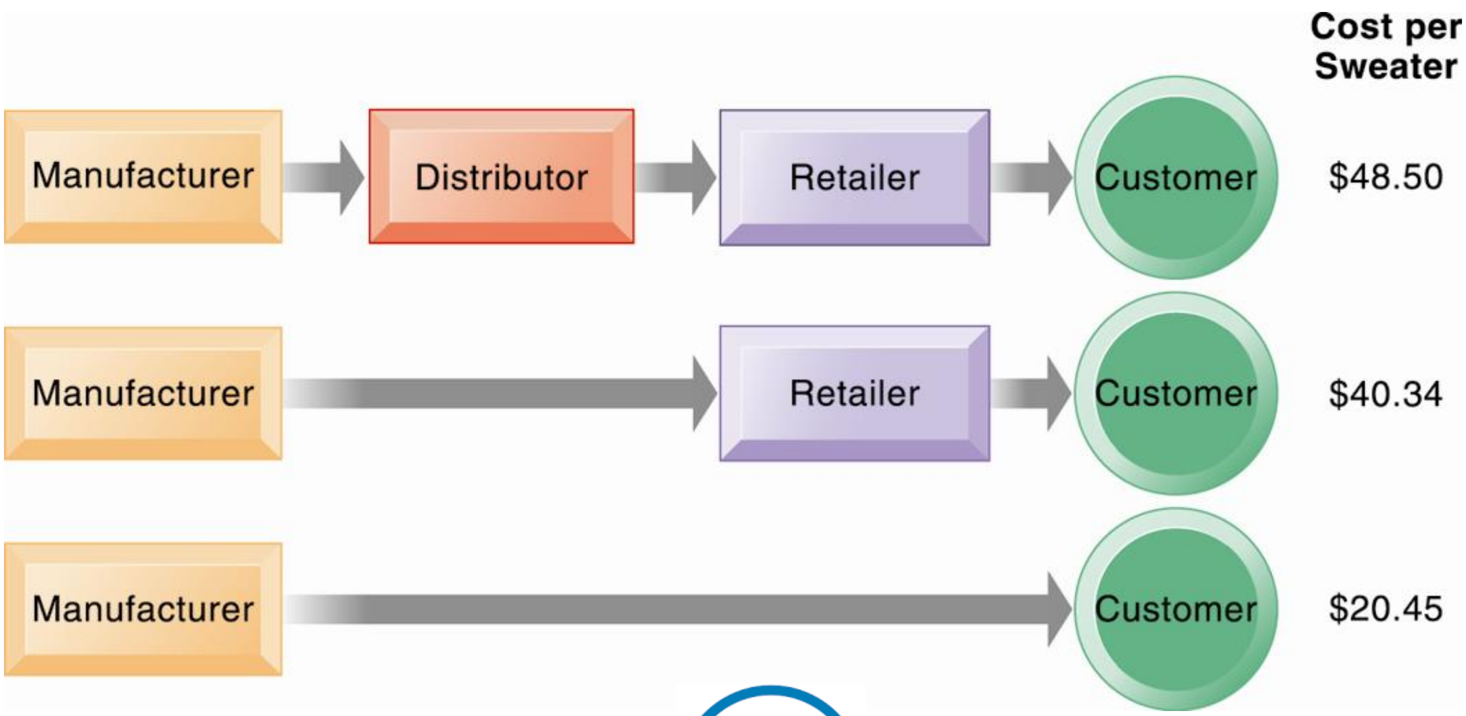
Source: Laudon and Laudon (2018)

Disintermediation

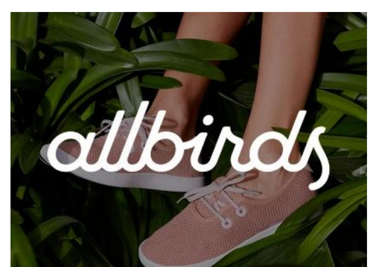
Disintermediation

- **The typical distribution channel has several intermediary layers, each of which adds to the final cost of a product.**
- **Removing layers (middlemen) lowers the final cost to the consumer.**

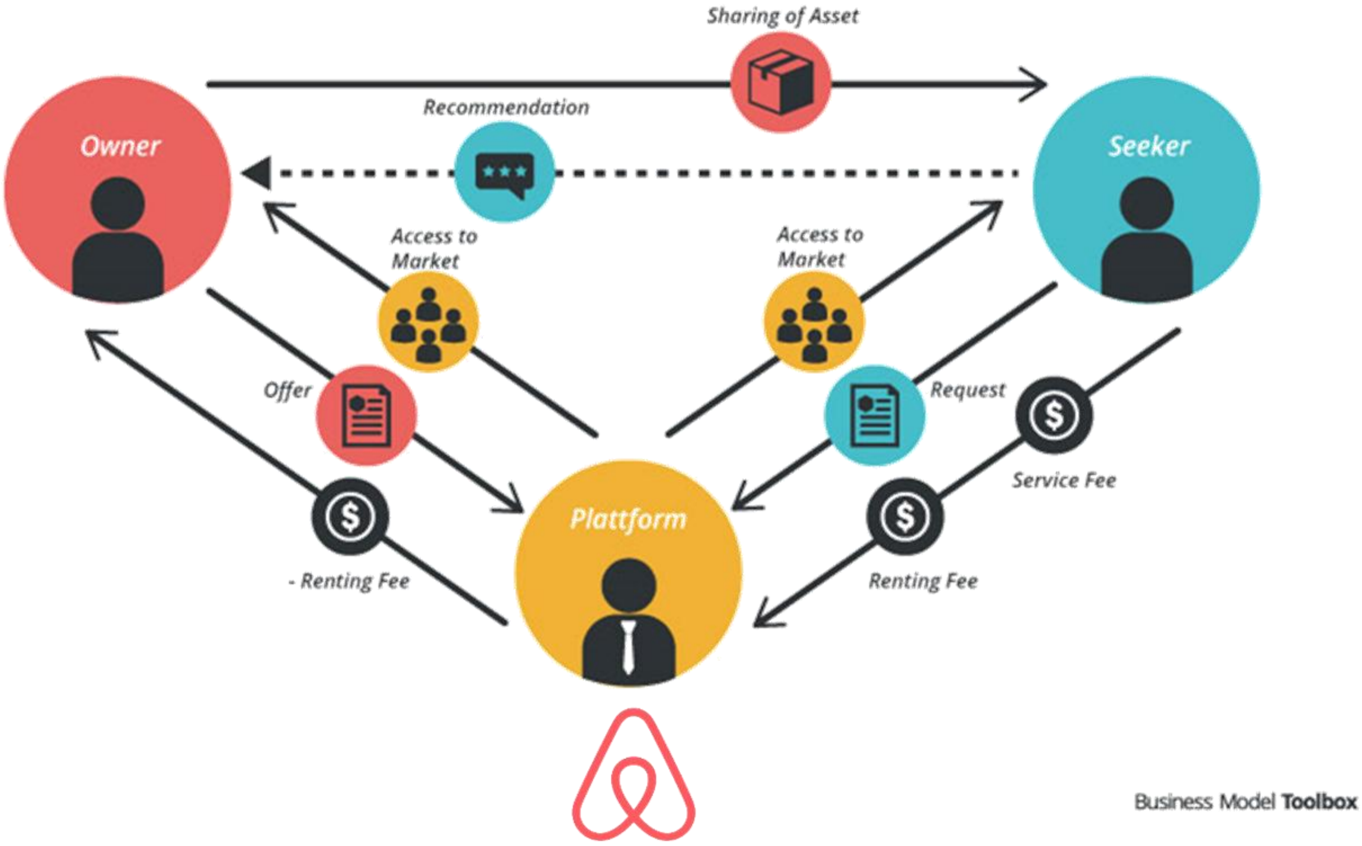


Direct-to-Consumer (D2C) companies sell products *directly* to customers online, bypassing traditional retail intermediaries.

- **Key Characteristics of D2C**
    - **Direct Communication:** D2C brands have a direct line of communication with their customers, allowing for better relationship building and customer service.
    - **Data-Driven:** Access to first-hand customer data, enabling a better understanding of customer needs and more strategic decision-making.
    - **Control:** D2C brands have more control over their messaging, marketing strategy, and fulfillment process.
    - **Digital-First:** Most D2C brands are digitally native, focusing on e-commerce.



However, the rise of platform businesses introduces new kinds of intermediation



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